Lesson notes · for you

Careers in cyber security

Cyber security · Level 4 · Lesson 4

Time	Materials
17 min	A phone or computer for the lab; Paper for the job card

Learning goals

- Describe several jobs in cyber security.
- Explain why permission comes first, and what to do if you find a weakness.
- Plan your own first steps.

Activity: Activity: Defender for a day

1. Warm-up (3 min): Read four short job descriptions and guess each job's name.
2. Lab (6 min): In the "Teach the machine" lab, teach the filter with safe and phishing examples. Write one tricky message it gets wrong, and say what example would fix it.
3. Sort (5 min): Sort six situation cards into "allowed" and "not allowed", and say why: a CTF for students; changing your own phone's settings; logging in to a classmate's account "to test it"; testing the campus network "to help" without asking; reporting a weakness privately; a club practice lab.
4. Plan (3 min): Write one step you will take this term.

Prepare six situation cards and four job descriptions on paper, one set per group.

Discussion

- Why is "just looking" without permission still a crime?
- What should you do if you notice a weakness by accident?
- Which of your university modules would help most in security, and why?

Check yourself

- One security job, and one first step I could take.

Try it this week

In the lab, train a phishing filter with examples, then find a message it gets wrong. Then pick one security job, find out what a day in it looks like, and tell a friend.

Worksheet

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Name _____ Class _____ Date _____

1 **While checking your own results, you see another student's details on the university's student portal by mistake. What's the ethical next step?**

- ◯ Look around to see what else shows up
- ◯ Tell the university's IT team privately, and go no further
- ◯ Post a warning on social media

2 **Match each job to what it does.**

Security analyst	• • • •	Makes sure people's data is treated fairly
Forensic investigator	• • • •	Tests systems, with written permission
Ethical hacker	• • • •	Works out what happened after an attack
Privacy officer	• • • •	Watches alarms and checks reported messages

3 **Looking around someone else's system without permission is legal, as long as you don't change anything.**

- ◯ True
- ◯ False

4 **Which are good first steps towards a security career for a student?**

- ☐ Join or start a computing club
- ☐ Enter capture-the-flag competitions made for students
- ☐ Take free courses and practise on labs built for practice
- ☐ Test your university's website to show off your skills
- ☐ Try classmates' passwords to show them they're weak

✎ **Which security job sounds most like you, and why? What is one thing you could do this term to find out more?**